Xi'an Jiaotong-Liverpool University
西交利物浦大学

XJTLU Entrepreneur College (Taicang) Cover Sheet

Module code and Title	**DTS410TC Fundamentals of Artificial Intelligence**
School Title	**School of AI and Advanced Computing**
Assignment Title	**Coursework2**
Submission Deadline	**07/Jun/2026 23:59**
Final Word Count	
If you agree to let the university use your work anonymously for teaching and learning purposes, please type **"yes"** here.	

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Scoring – For Tutor Use	
Student ID	

Stage of Marking	**Marker Code**	**Learning Outcomes Achieved (F/P/M/D) (please modify as appropriate)**				**Final Score**
		A	**B**	**C**	**D**	
1st Marker – red pen						
Moderation – green pen	**IM Initials**		The original mark has been accepted by the moderator (please circle as appropriate):			Y / N
			Data entry and score calculation have been checked by another tutor (please circle):			Y
2nd Marker if needed – green pen						
For Academic Office Use			**Possible Academic Infringement (please tick as appropriate)**			

Date Received	**Days late**	**Late Penalty**			
			☐ **Category A**		Total Academic Infringement Penalty (A,B, C, D, E, Please modify where necessary) ____________________
			☐ **Category B**		
			☐ **Category C**		
			☐ **Category D**		
			☐ **Category E**		

School of Artificial Intelligence and Advanced Computing
Xi'an Jiaotong-Liverpool University

DTS410TC Generative Artificial Intelligence

Coursework 2: Training and Analysis of GAN Variants for Face Image Generation

Due: 07/Jun/2026 23:59
Weight: 35%
Maximum score: 100 points (group work)

Learning Outcomes Assessed

- C. Expertly evaluate and appraise the performance of generative models using appropriate metrics and visualisation techniques.
- D. Collaborate systematically and creatively within teams in order to deal with complex generative AI issues that require communication and teamwork.

Team Policy

Students need to form a team of up to two members. All team members' information must be submitted via LMO no later than 5 April at 23:59. Students who do not submit their team information by the deadline will be randomly assigned to a team. Once teams have been confirmed, no further changes will be permitted.

Late policy

5% of the total marks available for the assessment shall be deducted from the assessment mark for each working day after the submission date, up to a maximum of five working days

Coding Policy

- **Programming Language:** All code must be written in Python with PyTorch.
- **Code Structure:** The code must be well-structured, readable, and properly documented. Comments should be included for each function and section, explaining its purpose and functionality.
- **Code Validation:** Ensure the code is fully executable without errors.

Avoid Plagiarism

- Do ***not*** submit work from other teams.
- Do ***not*** share code/work with other teams.
- Do ***not*** use open-source code as it is or without proper reference.

Risks

- Please read the coursework instructions and requirements carefully. Not following these instructions and requirements may result in a loss of marks.
- The assignment must be submitted via Learning Mall. Only electronic submission is accepted and no hard copy submission.
- All students must download their file and check that it is viewable after submission. Documents may become corrupted during the uploading process (e.g. due to slow internet connections). However, students are responsible for submitting a functional and correct file for assessments.
- Academic Integrity Policy is strictly followed.

Submission Requirements

Each group must submit a single zip file named `GroupName_Coursework2.zip` containing:

- **Group Report:** One group report in PDF format (maximum 30 pages), including the student IDs of all team members on the cover page. The report must include core implementation details, experimental analysis, and outputs (e.g., training curves, generated samples, FID results) so that results can be verified without rerunning the code.
- **Code:** A folder named `code/` containing all source code used in the project.
- **README:** A `README.md` file clearly explaining how to run the code, required libraries/dependencies, and installation/execution instructions.

Objective

In this coursework, your group will implement and train GANs from scratch to generate face images using the Align&Cropped version of the CelebA Dataset at 64×64 resolution. You will compare three GAN objectives (Vanilla GAN, LSGAN, and WGAN), evaluate generation quality using FID on the official test split, and analyze latent space interpolation behaviour. All variants must use the same generator/discriminator architectures, latent dimension (128), and image preprocessing.

Task 1: Implement the GAN Architecture [10 Marks]

- Implement a generator G that maps $z \in \mathbb{R}^{128}$ to a 64×64 RGB image using a convolutional architecture. Provide a clear architecture table (layers, channels, activations). [5 Marks]
- Implement a discriminator D suitable for 64×64 RGB images. Provide a clear architecture table (layers, channels, activations). [5 Marks]

Task 2: Implement and Compare Three GAN Objectives [20 Marks]

- **Vanilla GAN** Implement the standard GAN objective and describe the minimax formulation. [5 Marks]
- **Least-Squares GAN (LSGAN)** Implement the least-squares adversarial objective and explain how it modifies gradient behaviour. [7 Marks]
- **Wasserstein GAN (WGAN)** Implement the Wasserstein GAN objective with weight clipping. Clearly explain: (1) Why the sigmoid activation is removed; (2) The role of the Lipschitz constraint. [8 Marks]

Task 3: Training Dynamics, Sample Evolution, and Stability [25 Marks]

- For each objective, report training curves for generator and discriminator losses. Clearly comment on convergence behaviour and instability. [10 Marks]
- Provide qualitative sample grids at **at least three checkpoints** (early, mid, final training) for each objective. [10 Marks]
- Present a final comparison figure showing generated samples from all three objectives side-by-side. [5 Marks]

Task 4: FID Evaluation and Controlled Comparison [20 Marks]

FID is mandatory and must be computed using the official **test split** as the real-image reference.

- Clearly specify your FID protocol: number of generated samples (e.g., 5,000), preprocessing steps, and implementation tool. Use the same protocol for all objectives. [4 Marks]
- Report FID for Vanilla GAN, LSGAN, and WGAN in a structured comparison table. [8 Marks]
- Discuss whether FID rankings align with qualitative inspection. [8 Marks]

Task 5: Latent Space Interpolation and Representation Analysis [10 Marks]

- For each objective, perform linear interpolation between two randomly generated latent codes:

  $$z_\alpha = (1 - \alpha)z_1 + \alpha z_2, \quad \alpha \in [0, 1]$$

  Generate at least 8 intermediate samples per objective. [5 Marks]

- Compare interpolation smoothness, semantic transitions, and artifact behaviour across objectives. [5 Marks]

Task 6 Discussion and Reflection [15 Marks]

- Which objective achieved the best trade-off between stability, sample quality, and diversity? [5 Marks]

- Show one clear failure case from your GAN training (e.g., mode collapse, artifacts, unstable interpolation). Explain why this failure occurs and relate it to the objective or training dynamics. [5 Marks]

- Propose one technique that could improve the performance of all three GANs. Briefly explain why it would improve stability or FID. [5 Marks]